

Populating & Querying Database Documentation

1. Integrity Constraints & Data Validation

Describe how you ensured that each table in your database includes appropriate integrity constraints to enforce data accuracy and consistency, and how you performed data validation checks when inserting data. Include details on:

- Implementing primary key constraints to uniquely identify each record.
- Applying not null constraints to essential attributes.
- Enforcing unique constraints to ensure attribute distinctiveness.
- Setting check constraints to enforce specific conditions on attribute values.
- Maintaining referential integrity through foreign key constraints.
- Your description should outline the steps taken during the table creation process to ensure data integrity and accuracy, as well as the validation checks performed during data insertion.

Integrity Constraints

To guarantee accuracy, consistency, and relationship authenticity, integrity constraints were built into the design of each of the 17 tables in the research database.

Primary Keys:

Every entity table includes an **AUTO_INCREMENT** primary key (person_id, project_id, task_id, experiment_id, literature_id) to uniquely ID each record.

To avoid duplicate pairings, all relationship tables use composite primary keys made up of their foreign key fields.

NOT NULL Constraints:

In order to prevent the insertion of incomplete entries, essential characteristics including names, titles, status, email addresses, and dates were set as NOT NULL.

UNIQUE Constraints:

UNIQUE constraints were made on

- person.email
- department.abbreviation
- literature.doi

This prevents duplicate entries for important identifying info

CHECK Constraints

The following proper data ranges and values were enforced using CHECK constraints

- task.status limited to: Not Started, In Progress, Completed
- task.progress restricted to values between 0 and 100
- project.status
- experiment.status

These constraints make sure logical consistency is held inside our DB

Foreign Key Constraints

By using foreign key constraints that link back to their parent entity tables, all relationship tables keep referential integrity.

Through doing this orphan records are avoided, and all relationships are sure to refer to legit already existing records.

Data Validation During Insertion

The database was populated using a structured SQL dump file (db-v2.sql) imported into MySQL.

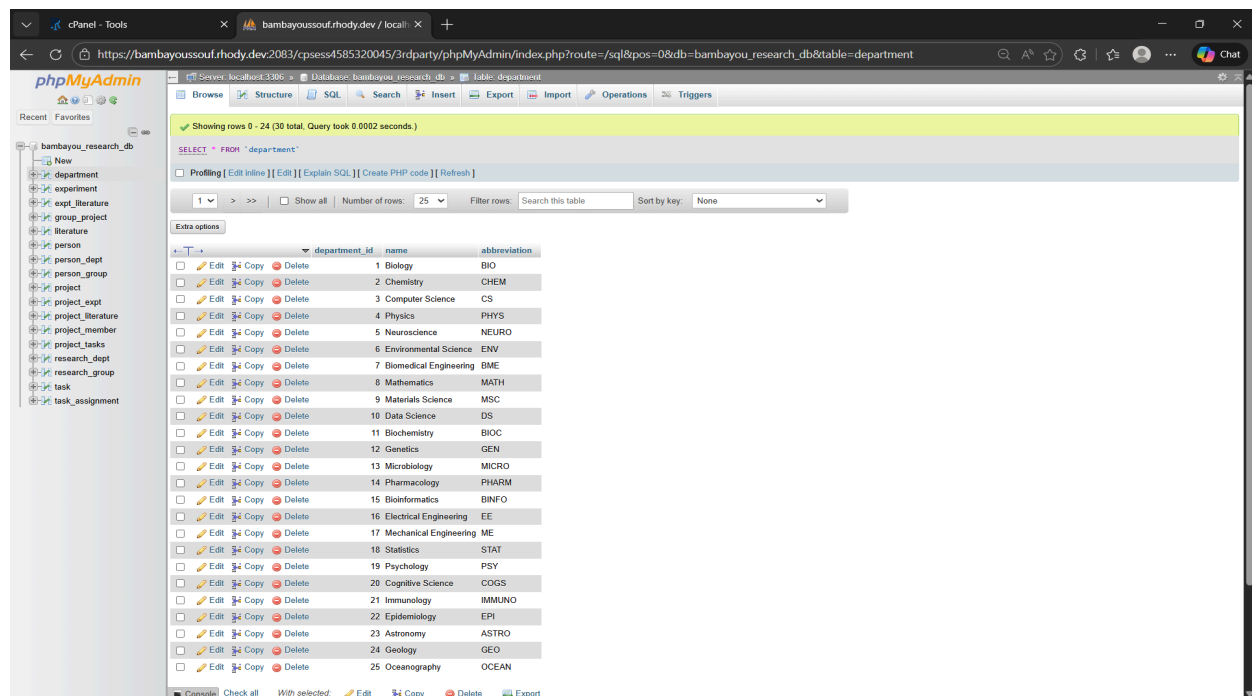
A redesigned dataset was created in order to better represent a real university research environment because the previous dataset lacked appropriate many-to-many relationships. The updated dataset assures that:

- People belong to multiple projects and research groups
- Tasks are assigned to multiple people
- Group level queries return varied, meaningful results rather than everyone showing the same flat count

2. Database Table Screenshots

Include screenshots of EACH table in your database. You can use the Browse tab and take a screenshot of the data that is displayed. This will demonstrate that you have successfully populated the database.

department



The screenshot shows the phpMyAdmin interface for the 'bambayou_research_db' database. The 'department' table is selected, and the 'Browse' tab is active. The table contains 25 rows of data, each representing a department with its ID, name, and abbreviation. The interface includes a sidebar with a tree view of the database structure, a top navigation bar with tabs like 'Browse', 'Structure', 'SQL', etc., and a main content area displaying the table data. The status bar at the bottom indicates 'Showing rows 0 - 24 (25 total. Query took 0.0002 seconds.)'.

department_id	name	abbreviation
1	Biology	BIO
2	Chemistry	CHEM
3	Computer Science	CS
4	Physics	PHYS
5	Neuroscience	NEURO
6	Environmental Science	ENV
7	Biomedical Engineering	BME
8	Mathematics	MATH
9	Materials Science	MSC
10	Data Science	DS
11	Biochemistry	BIOC
12	Genetics	GEN
13	Microbiology	MICRO
14	Pharmacology	PHARM
15	Bioinformatics	BINFO
16	Electrical Engineering	EE
17	Mechanical Engineering	ME
18	Statistics	STAT
19	Psychology	PSY
20	Cognitive Science	COGS
21	Immunology	IMMUNO
22	Epidemiology	EPI
23	Astronomy	ASTRO
24	Geology	GEO
25	Oceanography	OCEAN

experiment

Showing rows 0 - 24 (30 total, Query took 0.0002 seconds)

```
SELECT * FROM `experiment`
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

1 > >> Show all Number of rows: 25 Filter rows: Search this table Sort by key: None

Extra options

	experiment_id	title	objective	description	start_date	end_date	status
☐	Edit Copy Delete	1	CRISPR Transfection Trial 1	Optimize transfection efficiency	2023-02-01	2023-04-30	Completed
☐	Edit Copy Delete	2	Qubit Coherence Time Test	Measure T1 and T2 coherence times	2022-07-01	2022-09-30	Completed
☐	Edit Copy Delete	3	MRI Dataset Labeling	Create annotated training set	2022-03-15	2023-06-15	Completed
☐	Edit Copy Delete	4	Ice Core Sample Analysis	Extract climate proxies from ice cores	2021-10-01	2022-03-31	Completed
☐	Edit Copy Delete	5	LTP Induction in Slices	Induce and record LTP in brain slices	2022-08-01	2023-11-30	Completed
☐	Edit Copy Delete	6	Scaffold Porosity Test	Determine optimal pore size for cell infiltration	2023-01-01	2023-05-31	Completed
☐	Edit Copy Delete	7	Twitter Corpus Collection	Gather 2M tweets for analysis	2023-05-20	2023-07-20	Completed
☐	Edit Copy Delete	8	THz Source Calibration	Calibrate new THz emitter	2024-02-01	2024-04-30	Completed
☐	Edit Copy Delete	9	Fibril Cryo-EM Prep	Prepare amyloid samples for cryo-EM	2023-03-01	2023-05-01	Completed
☐	Edit Copy Delete	10	Kinase Binding Free Energy Calc	Compute binding affinities	2022-05-01	2022-10-31	Completed
☐	Edit Copy Delete	11	CAR-T Cytotoxicity Assay	Measure killing efficiency against HER2+ tumor cel...	2022-05-01	2022-07-31	Completed
☐	Edit Copy Delete	12	Variant Sequencing Run Batch 1	Sequence 500 SARS-CoV-2 isolates	2022-02-01	2022-05-31	Completed
☐	Edit Copy Delete	13	Graphene CVD Growth Optimization	Maximize monolayer coverage on Cu foil	2022-07-01	2023-10-31	Completed
☐	Edit Copy Delete	14	JVST Spectrum Extraction	Extract transmission spectra from JVST archive dat...	2022-09-01	2023-01-31	Completed
☐	Edit Copy Delete	15	fMRI Cognitive Task Protocol	Establish n-back task baseline in young adults	2023-10-01	2024-01-31	Completed
☐	Edit Copy Delete	16	Sediment Trap Deployment	Collect sinking particles at 2000 m depth	2021-06-01	2022-06-01	Completed
☐	Edit Copy Delete	17	Robot Arm Precision Calibration	Achieve sub-mm positional accuracy	2023-11-01	2024-02-28	Completed
☐	Edit Copy Delete	18	Ancient DNA Extraction Trial	Optimize aDNA recovery from degraded bone	2022-04-01	2022-07-31	Completed
☐	Edit Copy Delete	19	NO2 Sensor Cross-Validation	Validate electrochemical sensors against reference...	2022-08-01	2023-09-30	Completed
☐	Edit Copy Delete	20	Hydrogel Frequency Sweep	Characterize storage and loss moduli	2022-11-01	2023-02-28	Completed
☐	Edit Copy Delete	21	MRSA Whole Genome Sequencing	Generate reference genomes for 100 MRSA isolates	2022-02-01	2023-06-30	Completed
☐	Edit Copy Delete	22	Place Cell Firing Rate Simulation	Simulate theta-phase precession in model neurons	2022-09-01	2023-01-31	Completed
☐	Edit Copy Delete	23	Droplet Generation Rate Test	Achieve 1000 droplets/sec throughput	2023-06-01	2023-09-30	Completed
☐	Edit Copy Delete	24	Arctic Tern GPS Tagging	Deploy GPS loggers on 50 breeding individuals	2021-07-01	2021-08-31	Completed
☐	Edit Copy Delete	25	ICP-MS Isotope Measurement	Measure Ce, Nd, Sm isotope ratios in core samples	2022-10-01	2023-03-31	Completed

Console Check all With selected Edit Copy Delete Export

group_project

cPanel - Tools

bambayoussouf.rhody.dev/ local: /

https://bambayoussouf.rhody.dev/2083/cpsaes4585320045/3rdparty/phpMyAdmin/index.php?route=/sql&pos=0&db=bambayou_research_db&table=group_project

phpMyAdmin

Recent Favorites

bambayou_research_db

New

department

experiment

extt_literature

group_project

literature

person

person_dept

person_group

project

project_extt

project_literature

project_member

project_tasks

research_dept

research_group

task

task_assignment

Server: localhost:8888 > Database: bambayou_research_db > Table: group_project

Browse

Structure

SQL

Search

Insert

Export

Import

Operations

Triggers

Showing rows 0 - 24 (35 total, Query took 0.0002 seconds)

SELECT * FROM `group\_project`

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

1 > >>

Show all

Number of rows: 25

Filter rows: Search this table

Sort by key: None

Extra options

group_id

project_id

☐	Edit Copy Delete	1	1
☐	Edit Copy Delete	2	2
☐	Edit Copy Delete	3	3
☐	Edit Copy Delete	7	3
☐	Edit Copy Delete	4	4
☐	Edit Copy Delete	5	5
☐	Edit Copy Delete	6	6
☐	Edit Copy Delete	7	7
☐	Edit Copy Delete	8	8
☐	Edit Copy Delete	9	9
☐	Edit Copy Delete	10	10
☐	Edit Copy Delete	11	11
☐	Edit Copy Delete	28	11
☐	Edit Copy Delete	12	12
☐	Edit Copy Delete	13	13
☐	Edit Copy Delete	14	14
☐	Edit Copy Delete	3	15
☐	Edit Copy Delete	15	15
☐	Edit Copy Delete	16	16
☐	Edit Copy Delete	17	17
☐	Edit Copy Delete	1	18
☐	Edit Copy Delete	18	18
☐	Edit Copy Delete	19	19
☐	Edit Copy Delete	20	20
☐	Edit Copy Delete	21	21

Console

Check all 15

With selected

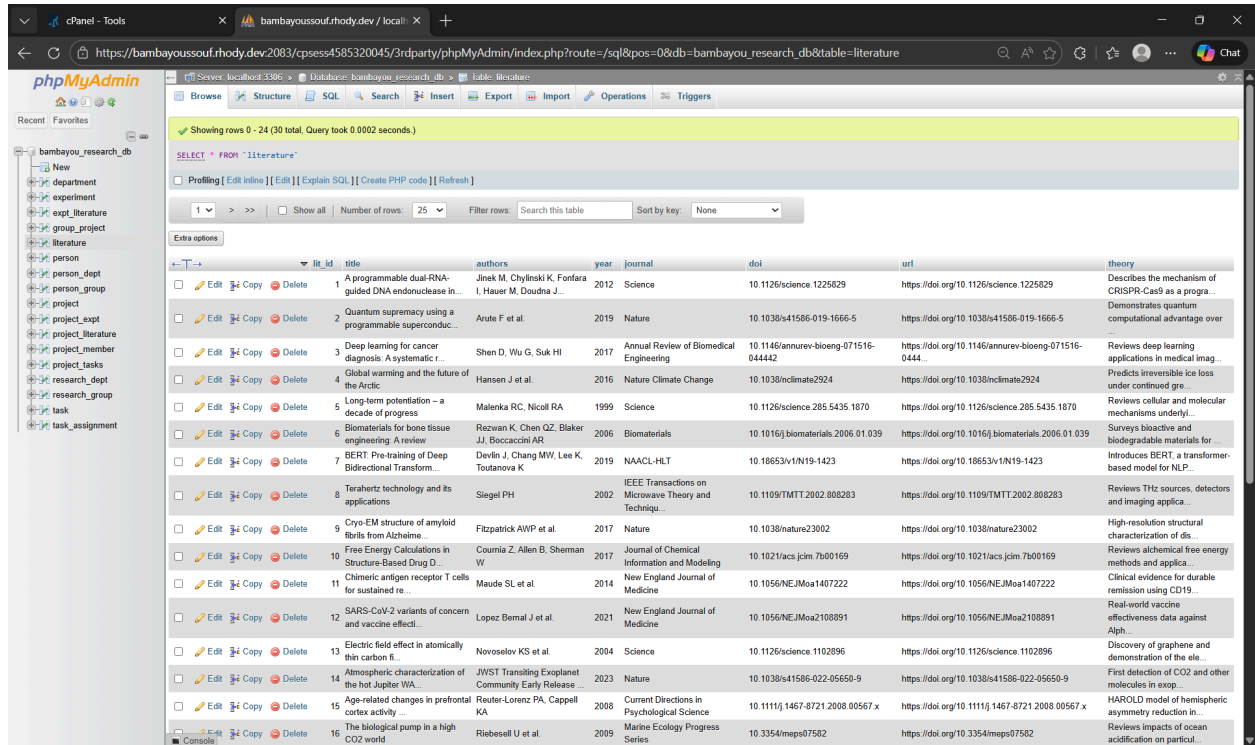
[Edit](#)

[Copy](#)

[Delete](#)

[Export](#)

literature



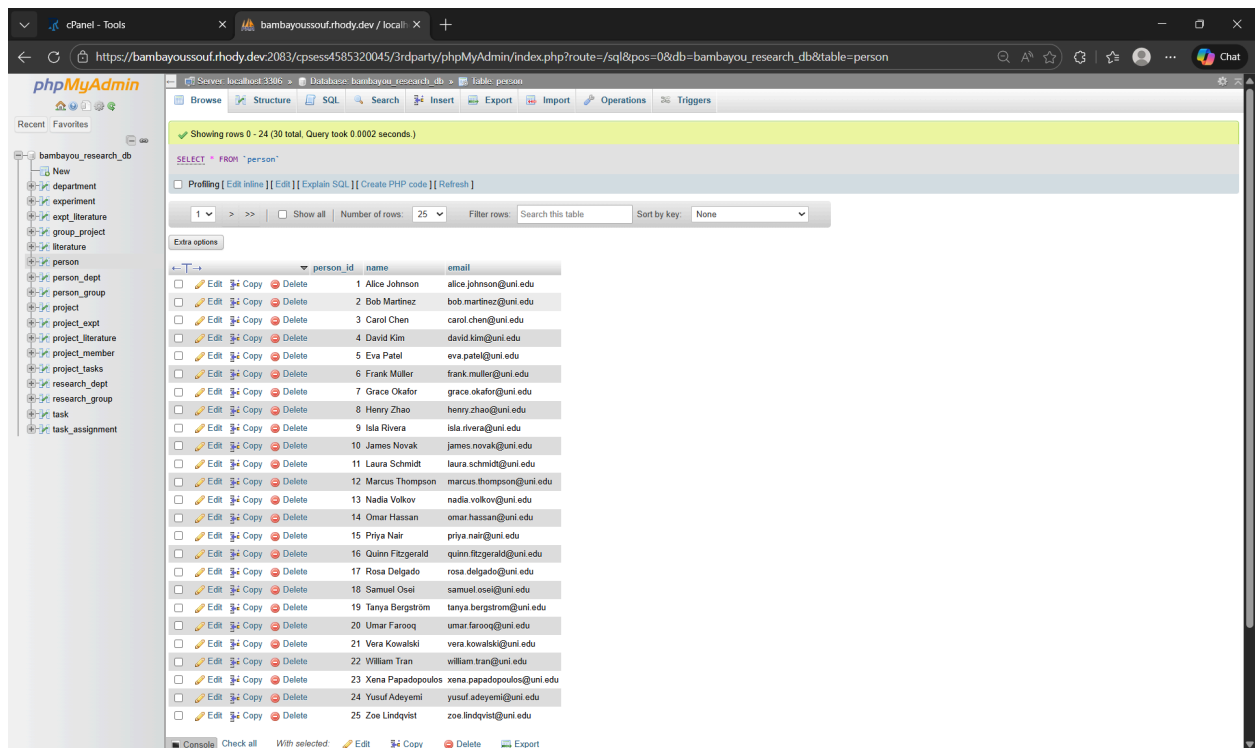
Showing rows 0 - 24 (30 total, Query took 0.0002 seconds)

```
SELECT * FROM `literature`
```

Extra options

	lit_id	title	authors	year	journal	doi	url	theory
☐	1	A programmable dual-RNA-guided DNA endonuclease in...	Jinek M, Chylinski K, Fonfara I, Hauer M, Doudna J...	2012	Science	10.1126/science.1225829	https://doi.org/10.1126/science.1225829	Describes the mechanism of CRISPR-Cas9 as a progra...
☐	2	Quantum supremacy using a programmable superconduct...	Arute F et al.	2019	Nature	10.1038/s41586-019-1666-5	https://doi.org/10.1038/s41586-019-1666-5	Demonstrates quantum computational advantage over ...
☐	3	Deep learning for cancer diagnosis: A systematic r...	Shan D, Wu G, Suk HI	2017	Annual Review of Biomedical Engineering	10.1146/annurev-bioeng-071516-044442	https://doi.org/10.1146/annurev-bioeng-071516-0444...	Reviews deep learning applications in medical imag...
☐	4	Global warming and the future of the Arctic	Hansen J et al.	2016	Nature Climate Change	10.1038/nclimate2924	https://doi.org/10.1038/nclimate2924	Predicts irreversible ice loss under continued gra...
☐	5	Long-term potentiation - a decade of progress	Malenka RC, Nicoll RA	1999	Science	10.1126/science.285.5435.1870	https://doi.org/10.1126/science.285.5435.1870	Reviews cellular and molecular mechanisms underly...
☐	6	Biomaterials for bone tissue engineering: A review	Rezwani K, Chen QZ, Blaker JJ, Boccacini AR	2006	Biomaterials	10.1016/j.biomaterials.2006.01.039	https://doi.org/10.1016/j.biomaterials.2006.01.039	Surveys bioactive and biodegradable materials for ...
☐	7	BERT: Pre-training of Deep Bidirectional Transform...	Devlin J, Chang MW, Lee K, Toutanova K	2019	NAACL-HLT	10.18653/v1/N19-1423	https://doi.org/10.18653/v1/N19-1423	Introduces BERT, a transformer-based model for NLP...
☐	8	Terahertz technology and its applications	Siegel PH	2002	IEEE Transactions on Microwave Theory and Techniqu...	10.1109/TMTT.2002.808283	https://doi.org/10.1109/TMTT.2002.808283	Reviews THz sources, detectors and imaging applica...
☐	9	Cryo-EM structure of amyloid fibrils from Alzheimer...	Fitzpatrick AWP et al.	2017	Nature	10.1038/nature23002	https://doi.org/10.1038/nature23002	High-resolution structural characterization of dis...
☐	10	Free Energy Calculations in Structure-Based Drug D...	Cournia Z, Allen B, Sherman W	2017	Journal of Chemical Information and Modeling	10.1021/acs.jcim.7b00169	https://doi.org/10.1021/acs.jcim.7b00169	Reviews alchemical free energy methods and applica...
☐	11	Chimeric antigen receptor T cells for sustained re...	Maude SL et al.	2014	New England Journal of Medicine	10.1056/NEJMoa1407222	https://doi.org/10.1056/NEJMoa1407222	Clinical evidence for durable remission using CD19...
☐	12	SARS-CoV-2 variants of concern and vaccine effect...	Lopez Bernal J et al.	2021	New England Journal of Medicine	10.1056/NEJMoa2108891	https://doi.org/10.1056/NEJMoa2108891	Real-world vaccine effectiveness data against Alph...
☐	13	Electric field effect in atomically thin carbon fi...	Novoselov KS et al.	2004	Science	10.1126/science.1102896	https://doi.org/10.1126/science.1102896	Discovery of graphene and demonstration of the ele...
☐	14	Atmospheric characterization of the hot Jupiter WA...	JWST Transiting Exoplanet Community Early Release	2023	Nature	10.1038/s41586-022-05650-9	https://doi.org/10.1038/s41586-022-05650-9	First detection of CO2 and other molecules in exop...
☐	15	Age-related changes in prefrontal cortex activity ...	Reuter-Lorenz PA, Cappell KA	2008	Current Directions in Psychological Science	10.1111/j.1467-8721.2008.00567.x	https://doi.org/10.1111/j.1467-8721.2008.00567.x	HAROLD model of hemispheric asymmetry reduction in...
☐	16	The biological pump in a high CO2 world	Riebesell U et al.	2009	Marine Ecology Progress Series	10.3354/meps07582	https://doi.org/10.3354/meps07582	Reviews impacts of ocean acidification on particul...

person



Showing rows 0 - 24 (30 total, Query took 0.0002 seconds)

```
SELECT * FROM `person`
```

Extra options

	person_id	name	email
☐	1	Alice Johnson	alice.johnson@uni.edu
☐	2	Bob Martinez	bob.martinez@uni.edu
☐	3	Carol Chen	carol.chen@uni.edu
☐	4	David Kim	david.kim@uni.edu
☐	5	Eva Patel	eva.patel@uni.edu
☐	6	Frank Muller	frank.muller@uni.edu
☐	7	Grace Okafor	grace.okafor@uni.edu
☐	8	Henry Zhao	henry.zhao@uni.edu
☐	9	Isla Rivera	isla.rivera@uni.edu
☐	10	James Novak	james.novak@uni.edu
☐	11	Laura Schmidt	laura.schmidt@uni.edu
☐	12	Marcus Thompson	marcus.thompson@uni.edu
☐	13	Nadia Volkov	nadia.volkov@uni.edu
☐	14	Omar Hassan	omar.hassan@uni.edu
☐	15	Priya Nair	priya.nair@uni.edu
☐	16	Quinn Fitzgerald	quinn.fitzgerald@uni.edu
☐	17	Rosa Delgado	rosa.delgado@uni.edu
☐	18	Samuel Osei	samuel.osei@uni.edu
☐	19	Tanya Bergström	tanya.bergstrom@uni.edu
☐	20	Umar Farooq	umar.farooq@uni.edu
☐	21	Vera Kowalski	vera.kowalski@uni.edu
☐	22	William Tran	william.tran@uni.edu
☐	23	Xena Papadopoulos	xena.papadopoulos@uni.edu
☐	24	Yusuf Adeyemi	yusuf.adeyemi@uni.edu
☐	25	Zoe Lindqvist	zoe.lindqvist@uni.edu

project

Panel - Tools

bambayoussouf.rhody.dev/ local: +

https://bambayoussouf.rhody.dev/2083/cpses/4585320045/3rdparty/phpMyAdmin/index.php?route=sql&pos=0&db=bambayou_research_db&table=project

Search

☆

🔗

👤

...

Chat

phpMyAdmin

Recent

Favorites

bambayou_research_db

New

department

experiment

expt_literature

group_project

literature

person

person_dept

person_group

project

project_expt

project_literature

project_member

research_tasks

research_dept

research_group

task

task_assignment

Server: localhost:3306

Database: bambayou_research_db

Table: project

Browse

Structure

SQL

Search

Insert

Export

Import

Operations

Triggers

Showing rows 0 - 24 (30 total, Query took 0.0002 seconds.)

SELECT * FROM `project`

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

1 > >>

Show all

Number of rows: 25

Filter rows: Search this table

Sort by key: None

Extra options

⌕

project_id

title

description

start_date

end_date

status

☐	Edit	Copy	Delete	1	CRISPR Gene Editing Study	Investigating CRISPR-Cas9 efficacy in human cell L...	2023-01-15	2024-12-31	Active
☐	Edit	Copy	Delete	2	Quantum Error Correction	Developing algorithms to minimize quantum decohere...	2022-06-01	2025-05-31	Active
☐	Edit	Copy	Delete	3	AI Tumor Detection	CNN-based model for early-stage tumor detection in...	2023-03-01	2024-09-30	Completed
☐	Edit	Copy	Delete	4	Arctic Ice Melt Simulation	Modeling ice sheet dynamics under climate change s...	2021-09-01	2023-08-31	Completed
☐	Edit	Copy	Delete	5	Synaptic Plasticity Analysis	Studying long-term potentiation in hippocampal neu...	2023-07-01	NULL	Active
☐	Edit	Copy	Delete	6	Biodegradable Scaffold Design	Creating scaffolds for bone tissue engineering	2022-11-01	2024-10-31	Active
☐	Edit	Copy	Delete	7	Social Media Sentiment Mining	Large-scale NLP analysis of public opinion trends	2023-05-15	2024-05-14	Completed
☐	Edit	Copy	Delete	8	Terahertz Imaging System	Building a sub-millimeter wave imaging device	2024-01-01	2026-01-01	Active
☐	Edit	Copy	Delete	9	Amyloid Fibril Structure	Cryo-EM study of amyloid aggregation pathways	2023-02-01	2025-01-31	Active
☐	Edit	Copy	Delete	10	Drug-Receptor Binding Simulation	MD simulations for novel kinase inhibitors	2022-04-01	2024-03-31	Completed
☐	Edit	Copy	Delete	11	CAR-T Cell Optimization	Engineering chimeric antigen receptors for improve...	2023-04-01	2025-03-31	Active
☐	Edit	Copy	Delete	12	COVID-19 Variant Surveillance	Genomic epidemiology of emerging SARS-CoV-2 varian...	2022-01-01	2023-12-31	Completed
☐	Edit	Copy	Delete	13	Graphene Transistor Fabrication	Developing sub-10nm graphene field-effect transist...	2023-06-01	NULL	Active
☐	Edit	Copy	Delete	14	Exoplanet Atmosphere Analysis	Spectroscopic characterization of habitable zone p...	2022-08-01	2024-07-31	Active
☐	Edit	Copy	Delete	15	Working Memory and Aging	fMRI study of prefrontal cortex activity in aging ...	2023-09-01	2025-08-31	Active
☐	Edit	Copy	Delete	16	Deep Sea Carbon Flux Study	Measuring biological pump efficiency in abyssal zo...	2021-05-01	2023-04-30	Completed
☐	Edit	Copy	Delete	17	Laparoscopic Robot Arm v2	Second-generation autonomous laparoscopic surgical...	2023-10-01	2026-09-30	Active
☐	Edit	Copy	Delete	18	Human Migration Genomics	SNP-based analysis of ancient human migration patt...	2022-03-01	2024-02-28	Completed
☐	Edit	Copy	Delete	19	Urban Air Quality Monitoring	Dense sensor network for real-time urban pollutant...	2023-07-01	2025-06-30	Active
☐	Edit	Copy	Delete	20	Hydrogel Viscoelasticity Study	Rheological characterization of polyacrylamide hyd...	2022-10-01	2024-09-30	Active
☐	Edit	Copy	Delete	21	MRSA Resistance Mechanism Mapping	WGS-based identification of antibiotic resistance ...	2023-01-01	2024-12-31	Active
☐	Edit	Copy	Delete	22	Hippocampal Place Cell Simulation	Computational model of spatial navigation in CA1 n...	2022-07-01	2024-06-30	Active
☐	Edit	Copy	Delete	23	Droplet Microfluidics for Blood Typing	Encapsulation-based rapid blood group determinatio...	2023-05-01	2025-04-30	Active
☐	Edit	Copy	Delete	24	Migratory Bird Tracking Study	GPS telemetry and foraging behavior of Arctic tern...	2021-06-01	2023-05-31	Completed
☐	Edit	Copy	Delete	25	Rare Earth Isotope Profiling	Lanthanide isotope ratios as tracers in sedimentar...	2022-09-01	2024-08-31	Active

Console

Check all

With selected:

[Edit](#)

[Copy](#)

[Delete](#)

[Export](#)

research_group

phpMyAdmin

Recent Favorites

bambayou_research_db

New

department

experiment

extl_literature

group_project

literature

person

person_dept

person_group

project

project_extl

project_literature

project_member

project_tasks

research_dept

research_group

task

task_assignment

Server: localhost:8888 > Database: bambayou_research_db > Table: research_group

Browse

Structure

SQL

Search

Insert

Export

Import

Operations

Triggers

Showing rows 0 - 24 (30 total, Query took 0.0001 seconds)

SELECT * FROM `research\_group`

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

1 > >> ☐ Show all Number of rows: 25 Filter rows: Search this table Sort by key: None

Extra options

group_id

name

focus

☐

Edit

Copy

Delete

1

Genomics Lab

Study of genetic sequences and gene expression

☐

Edit

Copy

Delete

2

Quantum Dynamics Group

Quantum computing and particle interactions

☐

Edit

Copy

Delete

3

AI in Medicine

Applying machine learning to clinical diagnostics

☐

Edit

Copy

Delete

4

Climate Modeling Team

Simulation of climate and environmental change

☐

Edit

Copy

Delete

5

Neural Circuits Lab

Mapping and studying neural connectivity

☐

Edit

Copy

Delete

6

Biomaterials Research Unit

Development of biocompatible materials

☐

Edit

Copy

Delete

7

Data Mining Group

Extracting patterns from large-scale datasets

☐

Edit

Copy

Delete

8

Photonics Lab

Light-matter interaction and optical devices

☐

Edit

Copy

Delete

9

Structural Biology Group

Protein folding and molecular structures

☐

Edit

Copy

Delete

10

Computational Chemistry Team

Molecular simulation and drug design

☐

Edit

Copy

Delete

11

Immunotherapy Lab

Developing T-cell based cancer immunotherapies

☐

Edit

Copy

Delete

12

Epidemiology Modeling Group

Tracking and forecasting infectious disease spread

☐

Edit

Copy

Delete

13

Nanoelectronics Unit

Fabrication and characterization of nanoscale devi...

☐

Edit

Copy

Delete

14

Astrobiology Research Team

Investigating conditions for life beyond Earth

☐

Edit

Copy

Delete

15

Cognitive Neuroscience Lab

Brain imaging and higher cognitive function studie...

☐

Edit

Copy

Delete

16

Marine Biogeochemistry Group

Studying carbon and nitrogen cycles in ocean syste...

☐

Edit

Copy

Delete

17

Robotic Surgery Lab

Designing autonomous surgical assist robots

☐

Edit

Copy

Delete

18

Population Genetics Group

Analysis of genetic variation across human populat...

☐

Edit

Copy

Delete

19

Atmospheric Chemistry Lab

Measuring trace gases and aerosols in the atmosphe...

☐

Edit

Copy

Delete

20

Soft Matter Physics Group

Studying polymers, gels, and liquid crystal system...

☐

Edit

Copy

Delete

21

Infectious Disease Genomics

Whole-genome sequencing of bacterial and viral pat...

☐

Edit

Copy

Delete

22

Computational Neuroscience Lab

Modeling neural circuits with differential equatio...

☐

Edit

Copy

Delete

23

Microfluidics Engineering Group

Lab-on-chip devices for point-of-care diagnostics

☐

Edit

Copy

Delete

24

Behavioral Ecology Team

Studying animal behavior in natural environments

☐

Edit

Copy

Delete

25

Geochemistry Research Unit

Isotope tracing in geological formations

bambayoussouf.rhody.dev:2083/cpsess4585320045/3rdparty/.../index.php

Delete

Export

task

Showing rows 0 - 24 (25 total, Query took 0.0004 seconds)

```
SELECT * FROM `task`
```

Extra options

task_id	title	description	status	due_date	completed_date	progress
1	Design sgRNA sequences	Design 5 guide RNAs targeting BRCA1 locus	Completed	2023-02-28	2023-02-25	100
2	Set up dilution refrigerator	Install and cool down the dilution refrigerator	Completed	2022-07-15	2022-07-12	100
3	Train CNN baseline model	Train ResNet-50 on initial labeled MRI set	Completed	2023-07-01	2023-06-28	100
4	Write climate model report	Summarize findings from simulation runs	Completed	2023-06-30	2023-07-02	100
5	Prepare neuron staining protocol	Optimize immunofluorescence staining for NMDA rece...	In Progress	2024-01-15	NULL	60
6	3D print scaffold prototypes	Print 10 prototype scaffolds using PLA and HA blan...	Completed	2023-03-31	2023-03-29	100
7	Build sentiment classifier	Fine-tune BERT model on labeled tweet dataset	Completed	2023-09-01	2023-08-30	100
8	Align optical components	Align beam path for THz spectrometer setup	In Progress	2024-06-01	NULL	45
9	Collect cryo-EM images	Image 200 grid squares at 300 kV	In Progress	2024-02-01	NULL	70
10	Validate docking poses	Cross-check MD poses with experimental IC50 data	Completed	2022-09-30	2022-09-28	100
11	Clone CAR construct into lentiviral vector	Gibson assembly of CD19-targeting CAR into pLenti...	Completed	2023-04-30	2023-04-28	100
12	Build variant phylogenetic tree	Maximum likelihood tree of 500 sequenced isolates	Completed	2022-06-30	2022-07-01	100
13	Deposit graphene onto SiO2 substrate	Wet transfer of CVD graphene onto 300nm SiO2/Si wa...	Completed	2023-11-30	2023-11-25	100
14	Cross-match JWST targets with Gaia catalog	Astrometric cross-matching of 40 planet-hosting st...	Completed	2022-10-31	2022-10-29	100
15	Recruit fMRI participants aged 60+	Screen and enroll 30 older adult volunteers for ag...	In Progress	2024-05-01	NULL	55
16	Analyze sediment trap organic carbon content	CHN elemental analysis of 120 trap samples	Completed	2022-08-31	2022-09-02	100
17	Write robot kinematics controller	Implement inverse kinematics for 6-DOF surgical ar...	In Progress	2024-07-01	NULL	40
18	Genotype 200 ancient samples with SNP panel	Targeted enrichment and genotyping of 500k SNP arr...	Completed	2022-09-30	2022-09-28	100
19	Deploy 20 air quality sensor nodes	Install, configure, and network electrochemical se...	Completed	2023-08-31	2023-08-30	100
20	Perform SEM imaging of hydrogel cross-section	Freeze-dry and sputter-coat samples for SEM visual...	In Progress	2024-08-01	NULL	30
21	Annotate MRSA resistance genes with ResFinder	Automated resistance gene annotation pipeline for ...	Completed	2023-07-31	2023-07-29	100
22	Tune network hyperparameters for place cell model	Grid search over connection weights and noise leve...	Completed	2023-03-31	2023-03-30	100
23	Fabricate PDMS microfluidic chips	Soft lithography using SU-8 master molds	Completed	2023-07-31	2023-07-28	100
24	Process and filter GPS tracking data	Remove erroneous fixes and interpolate gaps in tel...	Completed	2021-10-31	2021-10-30	100
25	Prepare sediment core sub-samples for ICP-MS	Acid digestion of 80 core intervals in clean lab c...	Completed	2022-11-30	2022-11-28	100

person_group

Showing rows 0 - 24 (38 total, Query took 0.0002 seconds)

```
SELECT * FROM `person_group`
```

Extra options

person_id	group_id	role	join_date
1	1	Principal Investigator	2020-01-10
1	11	Collaborator	2023-06-01
2	2	Principal Investigator	2019-06-01
3	3	Lead Data Scientist	2021-03-15
3	7	Data Consultant	2023-08-01
4	4	Senior Researcher	2020-09-01
5	5	Principal Investigator	2021-01-20
5	15	Visiting Researcher	2024-01-10
6	6	Researcher	2022-02-01
7	7	Research Associate	2022-05-10
8	8	Senior Researcher	2021-11-01
9	9	Postdoctoral Fellow	2023-01-05
10	10	Principal Investigator	2020-04-20
11	11	Principal Investigator	2021-04-01
11	28	Clinical Advisor	2023-09-01
12	12	Senior Researcher	2020-11-15
13	13	Lead Engineer	2022-06-01
14	14	Principal Investigator	2021-08-01
15	15	Principal Investigator	2022-09-01
15	26	fMRI Specialist	2024-02-01
16	16	Senior Researcher	2020-05-01
17	17	Lead Engineer	2022-10-01
18	1	Genomics Consultant	2022-10-01
18	18	Research Associate	2021-03-01
19	19	Postdoctoral Fellow	2022-07-01

project_expt

The screenshot shows the phpMyAdmin interface for the 'bambayou_research_db' database. The 'project_expt' table is selected, displaying 25 rows. The table has two columns: 'project_id' and 'experiment_id'. The data is as follows:

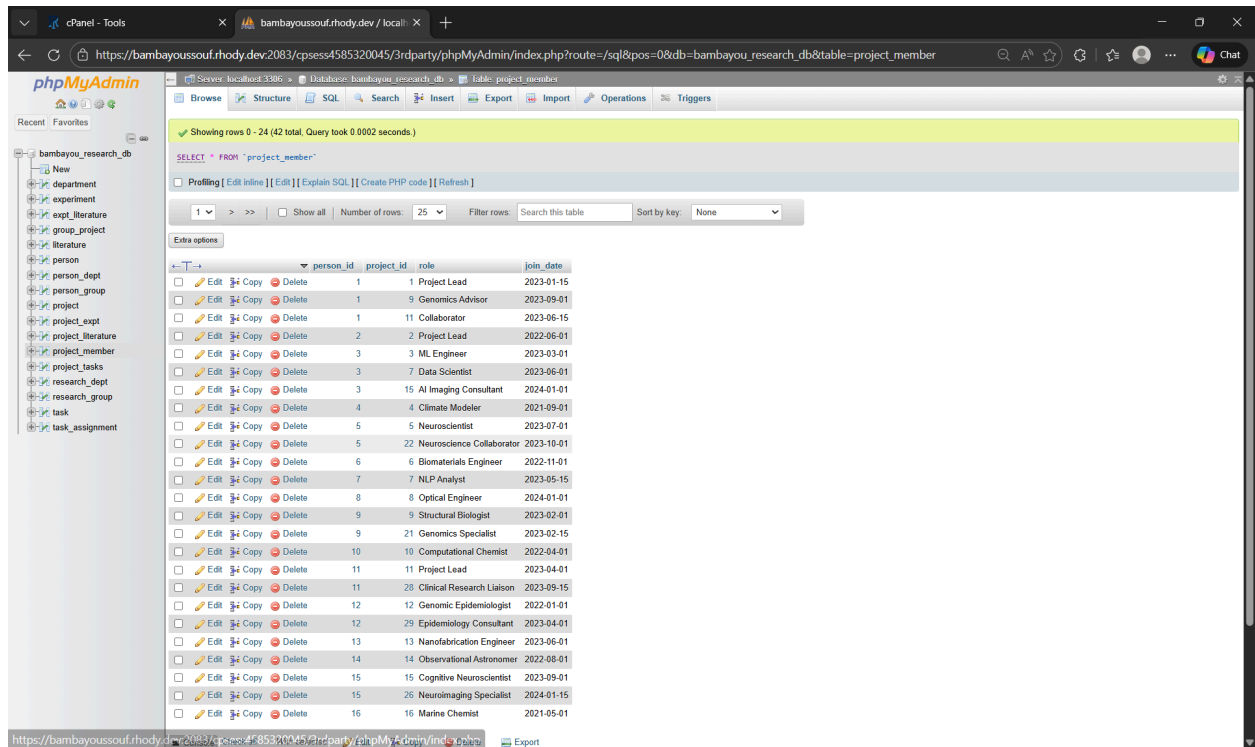
project_id	experiment_id
1	1
2	2
3	3
4	4
5	5
6	6
7	7
8	8
9	9
10	10
11	11
12	12
13	13
14	14
15	15
16	16
17	17
18	18
19	19
20	20
21	21
22	22
23	23
24	24
25	25

project_literature

The screenshot shows the phpMyAdmin interface for the 'bambayou_research_db' database. The 'project_literature' table is selected, displaying 20 rows. The table has two columns: 'project_id' and 'lit_id'. The data is as follows:

project_id	lit_id
1	1
11	1
2	2
3	3
4	4
5	5
22	5
6	6
3	7
7	7
8	8
9	9
10	10
1	11
11	11
28	11
12	12
13	13
14	14
15	15
16	16
17	17
18	18
19	19
20	20

project_member



Showing rows 0 - 24 (42 total, Query took 0.0002 seconds)

```
SELECT * FROM `project_member`
```

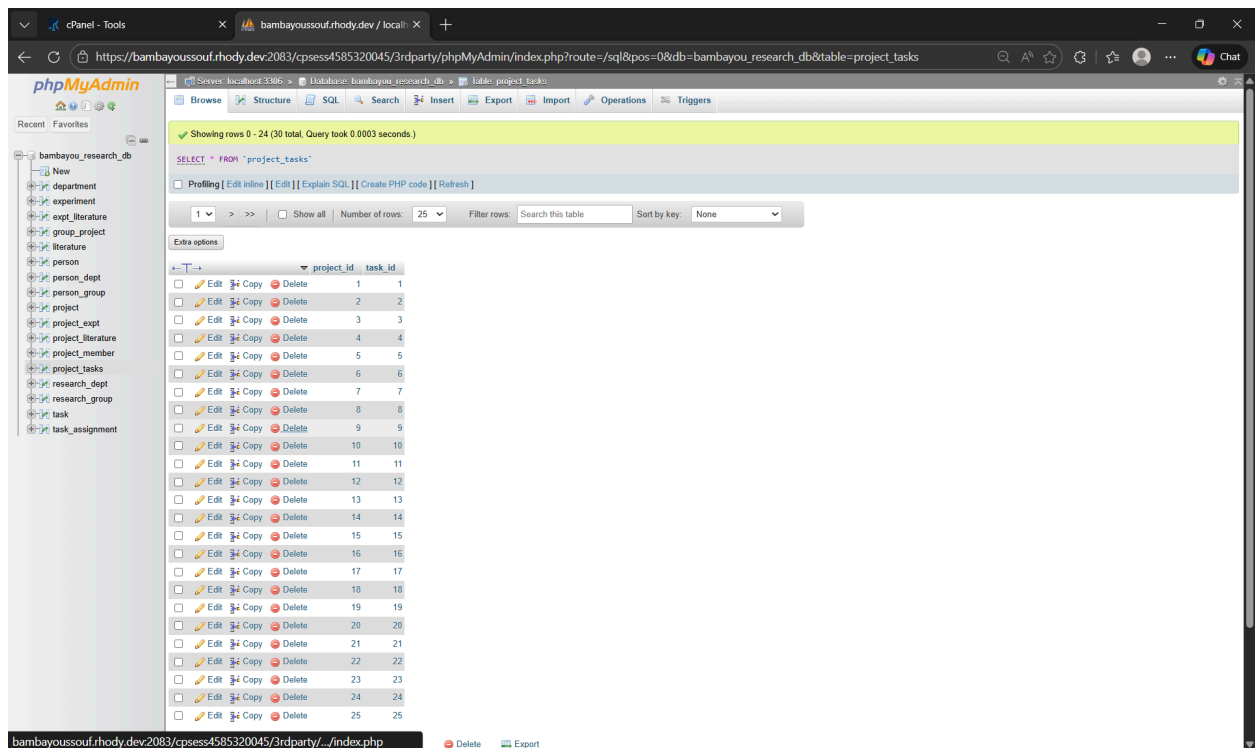
Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

1 > >> Show all Number of rows: 25 Filter rows: Search this table Sort by key: None

Extra options

	person_id	project_id	role	join_date
☐ Edit ☐ Copy ☐ Delete	1	1	Project Lead	2023-01-15
☐ Edit ☐ Copy ☐ Delete	1	9	Genomics Advisor	2023-09-01
☐ Edit ☐ Copy ☐ Delete	1	11	Collaborator	2023-06-15
☐ Edit ☐ Copy ☐ Delete	2	2	Project Lead	2022-06-01
☐ Edit ☐ Copy ☐ Delete	3	3	ML Engineer	2023-03-01
☐ Edit ☐ Copy ☐ Delete	3	7	Data Scientist	2023-06-01
☐ Edit ☐ Copy ☐ Delete	3	15	AI Imaging Consultant	2024-01-01
☐ Edit ☐ Copy ☐ Delete	4	4	Climate Modeler	2021-09-01
☐ Edit ☐ Copy ☐ Delete	5	5	Neuroscientist	2023-07-01
☐ Edit ☐ Copy ☐ Delete	5	22	Neuroscience Collaborator	2023-10-01
☐ Edit ☐ Copy ☐ Delete	6	6	Biomaterials Engineer	2022-11-01
☐ Edit ☐ Copy ☐ Delete	7	7	NLP Analyst	2023-05-15
☐ Edit ☐ Copy ☐ Delete	8	8	Optical Engineer	2024-01-01
☐ Edit ☐ Copy ☐ Delete	9	9	Structural Biologist	2023-02-01
☐ Edit ☐ Copy ☐ Delete	9	21	Genomics Specialist	2023-02-15
☐ Edit ☐ Copy ☐ Delete	10	10	Computational Chemist	2022-04-01
☐ Edit ☐ Copy ☐ Delete	11	11	Project Lead	2023-04-01
☐ Edit ☐ Copy ☐ Delete	11	28	Clinical Research Liaison	2023-09-15
☐ Edit ☐ Copy ☐ Delete	12	12	Genomic Epidemiologist	2022-01-01
☐ Edit ☐ Copy ☐ Delete	12	29	Epidemiology Consultant	2023-04-01
☐ Edit ☐ Copy ☐ Delete	13	13	Nanofabrication Engineer	2023-06-01
☐ Edit ☐ Copy ☐ Delete	14	14	Observational Astronomer	2022-08-01
☐ Edit ☐ Copy ☐ Delete	15	15	Cognitive Neuroscientist	2023-09-01
☐ Edit ☐ Copy ☐ Delete	15	26	Neuroimaging Specialist	2024-01-15
☐ Edit ☐ Copy ☐ Delete	16	16	Marine Chemist	2021-05-01

project_tasks



Showing rows 0 - 24 (30 total, Query took 0.0003 seconds)

```
SELECT * FROM `project_tasks`
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

1 > >> Show all Number of rows: 25 Filter rows: Search this table Sort by key: None

Extra options

	project_id	task_id
☐ Edit ☐ Copy ☐ Delete	1	1
☐ Edit ☐ Copy ☐ Delete	2	2
☐ Edit ☐ Copy ☐ Delete	3	3
☐ Edit ☐ Copy ☐ Delete	4	4
☐ Edit ☐ Copy ☐ Delete	5	5
☐ Edit ☐ Copy ☐ Delete	6	6
☐ Edit ☐ Copy ☐ Delete	7	7
☐ Edit ☐ Copy ☐ Delete	8	8
☐ Edit ☐ Copy ☐ Delete	9	9
☐ Edit ☐ Copy ☐ Delete	10	10
☐ Edit ☐ Copy ☐ Delete	11	11
☐ Edit ☐ Copy ☐ Delete	12	12
☐ Edit ☐ Copy ☐ Delete	13	13
☐ Edit ☐ Copy ☐ Delete	14	14
☐ Edit ☐ Copy ☐ Delete	15	15
☐ Edit ☐ Copy ☐ Delete	16	16
☐ Edit ☐ Copy ☐ Delete	17	17
☐ Edit ☐ Copy ☐ Delete	18	18
☐ Edit ☐ Copy ☐ Delete	19	19
☐ Edit ☐ Copy ☐ Delete	20	20
☐ Edit ☐ Copy ☐ Delete	21	21
☐ Edit ☐ Copy ☐ Delete	22	22
☐ Edit ☐ Copy ☐ Delete	23	23
☐ Edit ☐ Copy ☐ Delete	24	24
☐ Edit ☐ Copy ☐ Delete	25	25

research_dept

Showing rows 0 - 24 (24 total, Query took 0.0001 seconds)

```
SELECT * FROM `research_dept`
```

Extra options

	group_id	department_id
☐	1	1
☐	9	1
☐	24	1
☐	10	2
☐	30	2
☐	3	3
☐	2	4
☐	8	4
☐	5	5
☐	4	6
☐	19	6
☐	6	7
☐	23	7
☐	15	9
☐	7	10
☐	22	10
☐	11	11
☐	21	11
☐	28	11
☐	18	12
☐	14	13
☐	25	14
☐	16	15
☐	20	16
☐	17	17

task_assignment

Showing rows 0 - 24 (24 total, Query took 0.0001 seconds)

```
SELECT * FROM `task_assignment`
```

Extra options

	person_id	task_id	assigned_date
☐	1	1	2023-01-20
☐	1	9	2023-09-01
☐	1	11	2023-05-01
☐	2	2	2022-06-15
☐	2	8	2024-01-20
☐	3	3	2023-03-10
☐	3	7	2023-06-01
☐	3	15	2024-01-05
☐	4	4	2023-05-01
☐	5	5	2023-07-10
☐	6	6	2022-12-01
☐	7	7	2023-05-20
☐	8	8	2024-01-15
☐	9	9	2023-02-10
☐	9	21	2023-01-15
☐	10	10	2022-04-10
☐	11	11	2023-04-05
☐	11	28	2022-06-01
☐	12	12	2022-01-10
☐	12	29	2023-04-01
☐	13	13	2023-06-10
☐	14	14	2022-08-15
☐	15	15	2023-09-10
☐	15	26	2023-12-01
☐	16	16	2021-05-10

expt_literature

The screenshot shows the phpMyAdmin interface for the 'bambayou_research_db' database. The 'expt_literature' table is selected, and its structure is displayed. The table has two columns: 'experiment_id' and 'lit_id'. The data is shown in a table with 25 rows, each containing an 'experiment_id' and a corresponding 'lit_id'.

experiment_id	lit_id
1	1
2	2
3	3
4	4
5	5
6	6
7	7
8	8
9	9
10	10
11	11
12	12
13	13
14	14
15	15
16	16
17	17
18	18
19	19
20	20
21	21
22	22
23	23
24	24
25	25

https://bambayoussouf.rhody.dev:2083/cpsess4585320045/3rdparty/phpMyAdmin/index.php?route=/sql&pos=0&db=bambayou_research_db&table=expt_literature

person_dept

The screenshot shows the phpMyAdmin interface for the 'bambayou_research_db' database. The 'person_dept' table is selected, and its structure is displayed. The table has two columns: 'person_id' and 'department_id'. The data is shown in a table with 24 rows, each containing a 'person_id' and a corresponding 'department_id'.

person_id	department_id
1	1
9	1
10	2
30	2
3	3
2	4
8	4
5	5
4	6
19	6
26	6
6	7
15	9
7	10
23	10
25	10
11	11
12	12
18	12
14	13
21	13
24	14
16	15
20	16
29	16

https://bambayoussouf.rhody.dev:2083/cpsess4585320045/3rdparty/phpMyAdmin/index.php?route=/sql&pos=0&db=bambayou_research_db&table=person_dept

3. SQL Queries & Insights

Five SQL queries were written to extract meaningful insights from the database. Each query serves a distinct purpose and uses different SQL clauses and operators including:

- A query leveraging SELECT, FROM, and WHERE clauses with comparison and logical operators to filter rows based on specific conditions.
- A query employing arithmetic operators (+, -, *, /) to perform calculations within the queries.
- A query performing join expressions to combine data from multiple tables.
- A query using the GROUP BY clause and applying aggregate functions to perform group-level calculations.
- Each SQL query is documented with its intended purpose, expected results, and the actual SQL query written.
- The results of each SQL query are included showcasing the insights extracted from the database. Screenshots of the tables displaying the query results are provided.

Query 1 – Filtering In Progress Tasks Below 50%

Purpose:

Identify tasks that are in progress but less than halfway complete, allowing research groups to prioritize unfinished work.

SQL Query:

```
-- Query 1-> SELECT, FROM, WHERE with comparison and logical operators

-- "Which tasks are in progress with less than 50% completion?"

SELECT task_id, title, status, progress, due_date
FROM task
WHERE status = 'In Progress'
AND progress < 50
ORDER BY progress ASC;
```

Expected Result:

A table of task IDs, titles, progress percentages, and due dates for tasks below 50 percent and in progress.

Result:

task_id	title	status	progress	due_date
20	Perform SEM imaging of hydrogel cross-section	In Progress	30	2024-08-01
17	Write robot kinematics controller	In Progress	40	2024-07-01
8	Align optical components	In Progress	45	2024-06-01

Query 2 – Calculating Remaining Work and Days Until Due

Purpose:

Determine how much work remains on each in-progress task and how many days are left until the task is due. This helps track deadlines and task completion progress.

SQL Query:

```
-- Query 2: Arithmetic operators

-- "How much work remains on each in-progress task?"

SELECT title, progress,

       (100 - progress) AS percent_remaining,

       DATEDIFF(due_date, CURDATE()) AS days_until_due

FROM task

WHERE status = 'In Progress'

ORDER BY progress ASC;
```

Expected Result:

A table listing task titles, current progress, percent of work remaining, and number of days until each task's due date.

Result:

-- Query 1-> SELECT, FROM, WHERE wi...		-- Query 2: Arithmetic operators -- "Ho...		-- Query 3: JOIN across multiple tables	
title	progress	percent_remai...	days_until_due		
abc Filter...	abc Filter...	abc Filter...	abc Filter...		
Perform SEM imaging of hydrogel cross-section	30	70	-568		
Write robot kinematics controller	40	60	-599		
Align optical components	45	55	-629		
Optimize yeast fermentation media composition	50	50	-721		
Recruit fMRI participants aged 60+	55	45	-660		
Prepare neuron staining protocol	60	40	-767		
Preprocess EEG data with ICA	65	35	-629		
Collect cryo-EM images	70	30	-750		

Query 3 – Project Members and Their Roles

Purpose:

Retrieve the names, emails, and roles of all members associated with the project (in this case “CRISPR Gene Editing Study”). This allows tracking of team composition and responsibilities.

SQL Query:

```
-- Query 3: JOIN across multiple tables

-- "Who are the members of the CRISPR Gene Editing Study and what are their roles?"

SELECT p.name, p.email, pm.role
FROM person p
INNER JOIN project_member pm ON p.person_id = pm.person_id
INNER JOIN project pr ON pm.project_id = pr.project_id
WHERE pr.title = 'CRISPR Gene Editing Study';
```

Expected Result:

A table listing each person involved in the project along with their email and role in the project.

Result:

name	email	role	
abc Filter...	abc Filter...	abc Filter...	
Alice Johnson	alice.johnson@uni.edu	Project Lead	
Samuel Osei	samuel.osei@uni.edu	Bioinformatics Support	

Query 4 – Task Count & Average Progress for Each Person

Purpose:

Show how many tasks each person is assigned and their average progress across those tasks. This helps in evaluating performance and assigning tasks.

SQL Query:

```
-- Query 4: GROUP BY with aggregate functions

-- "How many tasks does each person have, and what is their average progress?"

SELECT p.name,

       COUNT(ta.task_id) AS total_tasks,

       AVG(t.progress) AS avg_progress

FROM person p

JOIN task_assignment ta ON p.person_id = ta.person_id

JOIN task t ON ta.task_id = t.task_id

GROUP BY p.person_id, p.name

ORDER BY total_tasks DESC;
```

Expected Result:

A table with each person's name, total number of tasks assigned, and their average task completion percentage.

Result:

name	total_tasks	avg_progress
abc Filter...	abc Filter...	abc Filter...
Samuel Osei	3	100.0000
Carol Chen	3	85.0000
Alice Johnson	3	90.0000
Isla Rivera	2	85.0000
Bob Martinez	2	72.5000
Priya Nair	2	60.0000
William Tran	2	80.0000
Marcus Thompson	2	100.0000
Carlos Mendes	2	100.0000
Laura Schmidt	2	100.0000
Nadia Volkov	1	100.0000
James Novak	1	100.0000
Henry Zhao	1	45.0000
Grace Okafor	1	100.0000
Frank Müller	1	100.0000
Eva Patel	1	60.0000
Omar Hassan	1	100.0000
David Kim	1	100.0000
Quinn Fitzgerald	1	100.0000
Rosa Delgado	1	40.0000
Tanya Bergström	1	100.0000
Umar Farooq	1	30.0000
Vera Kowalski	1	100.0000
Xena Papadopoulos	1	100.0000
Yusuf Adeyemi	1	100.0000
Zoe Lindqvist	1	100.0000
Aaron Castillo	1	65.0000
Beatrice Fontaine	1	100.0000
Diana Sato	1	100.0000
Elias Andersen	1	50.0000

Query 5 – Literature Published After 2015 with Theory Content

Purpose:

To help with research evaluation and reference. The query filters literature entries to only display articles published after 2015 that contain theoretical content.

SQL Query:

```
-- Query 5: Multiple filter conditions

-- "Show all literature published after 2015 that has theory content."

SELECT lit_id, title, authors, year, journal

FROM literature

WHERE year > 2015

AND theory IS NOT NULL

ORDER BY year DESC;
```

Expected Result:

A table listing literature ID, title, authors, publication year, and journal for relevant papers.

Result:

lit_id	title	authors	year	journal
14	Atmospheric characterization of the hot Jupiter WASP-39b	JWST Transiting Exoplanet Community Early Release S...	2023	Nature
27	Highly efficient perovskite solar cells via improved carrier management	Kim M et al.	2022	Science
12	SARS-CoV-2 variants of concern and vaccine effectiveness	Lopez Bernal J et al.	2021	New England Journal of Medicine
22	Theta sequences are essential for internally generated hippocampal firing fields	Wang Y et al.	2020	Nature Neuroscience
2	Quantum supremacy using a programmable superconducting processor	Arute F et al.	2019	Nature
7	BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding	Devlin J, Chang MW, Lee K, Toutanova K	2019	NAACL-HLT
3	Deep learning for cancer diagnosis: A systematic review	Shen D, Wu G, Suk HI	2017	Annual Review of Biomedical Engineering
9	Cryo-EM structure of amyloid fibrils from Alzheimer disease brain	Fitzpatrick AWP et al.	2017	Nature
10	Free Energy Calculations in Structure-Based Drug Design	Cournia Z, Allen B, Sherman W	2017	Journal of Chemical Information and Modeling
4	Global warming and the future of the Arctic	Hansen J et al.	2016	Nature Climate Change
26	The bilingual brain: Flexibility and control in the human language system	Abutalebi J, Green DW	2016	Psychological Science in the Public Interest